COGNIZANT

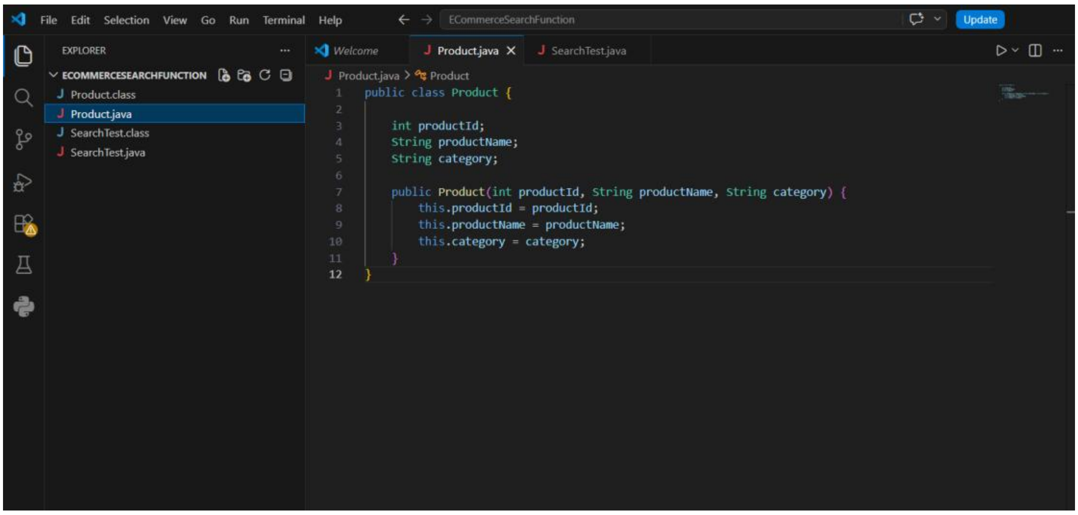
Mandatory Hands-on Exercises

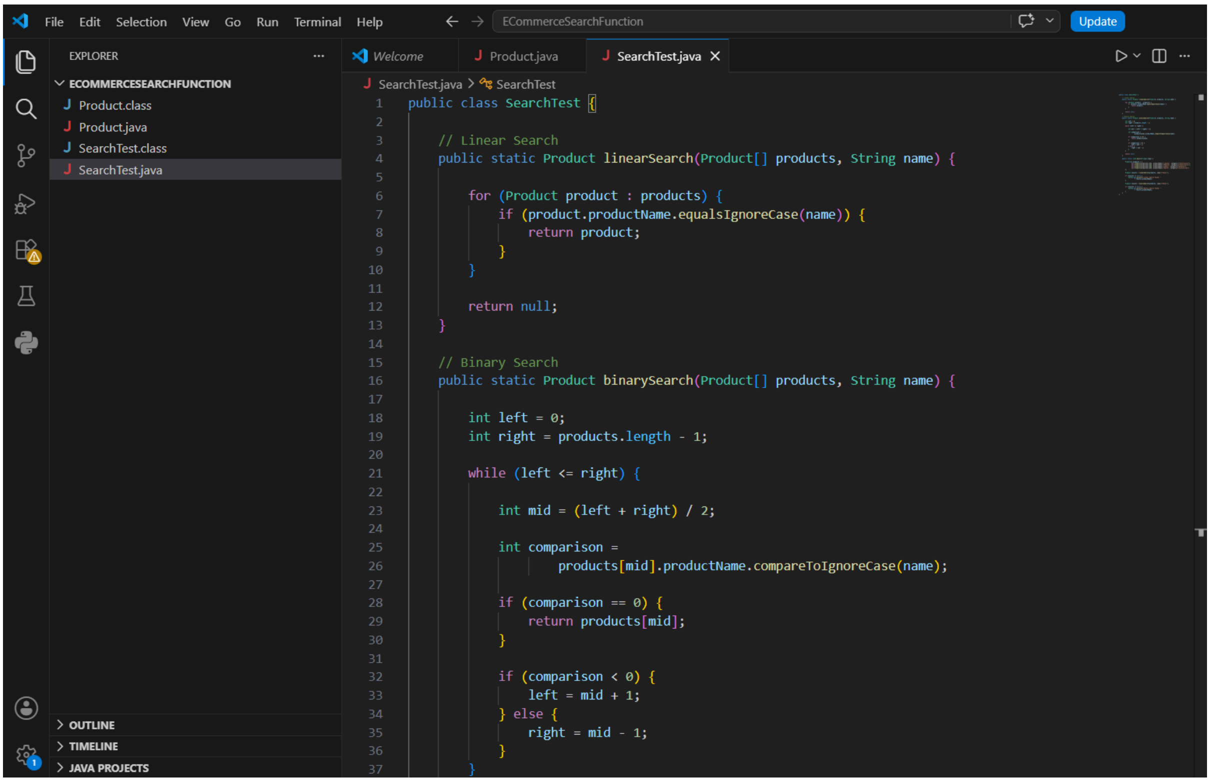
-KHUSHI AHI

TOPIC: Data Structures and Algorithms

<u>Exercise 2: E-commerce Platform Search Function</u>

CODE





```java

        return null;
    }

    Run | Debug
    public static void main(String[] args) {

        Product[] products = {
                new Product(productId: 101, productName: "Laptop", category: "Electronics"),
                new Product(productId: 102, productName: "Mobile", category: "Electronics"),
                new Product(productId: 103, productName: "Shoes", category: "Fashion"),
                new Product(productId: 104, productName: "Watch", category: "Accessories")
        };

        Product result1 = linearSearch(products, name: "Shoes");

        if (result1 != null) {
            System.out.println("Linear Search Found: "
                    + result1.productName);
        }

        Product result2 = binarySearch(products, name: "Shoes");

        if (result2 != null) {
            System.out.println("Binary Search Found: "
                    + result2.productName);
        }
    }
}
```

OUTPUT

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS cmd

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Khushi Ahi\OneDrive\Desktop\ECommerceSearchFunction>javac *.java

C:\Users\Khushi Ahi\OneDrive\Desktop\ECommerceSearchFunction>java SearchTest
Linear Search Found: Shoes
Binary Search Found: Shoes

C:\Users\Khushi Ahi\OneDrive\Desktop\ECommerceSearchFunction>
```

Exercise 7: Financial Forecasting

CODE

```java
public class Forecast {

    public static double futureValue(double currentValue,
                                     double growthRate,
                                     int years) {

        if (years == 0) {
            return currentValue;
        }

        return futureValue(
                currentValue * (1 + growthRate),
                growthRate,
                years - 1
        );
    }

    public static void main(String[] args) {

        double currentValue = 10000;
        double growthRate = 0.10;
        int years = 5;

        double result =
                futureValue(currentValue, growthRate, years);

        System.out.println("Future Value after "
                + years + " years = " + result);
    }
}
```

OUTPUT:

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Khushi Ahi\OneDrive\Desktop\FinancialForecasting>javac Forecast.java

C:\Users\Khushi Ahi\OneDrive\Desktop\FinancialForecasting>java Forecast
Future Value after 5 years = 16105.100000000008

C:\Users\Khushi Ahi\OneDrive\Desktop\FinancialForecasting>
```